

The Bourgain–Brezis critical Lorentz question cited in arXiv:1312.5823 was answered by arXiv:2606.12891

Literature-status packet for run `fa_banach_001`

11 August 2026

Status: literature already answered. Giga–Xiang cite Bourgain–Brezis Open Problem 1 and say that the divergence-free, curl-in- L^1 case was unknown. Goodman–Hernández–Spector explicitly identify the same question and prove the endpoint $L^{3/2,1}(\mathbb{T}^3)$ estimate. This is a retrieval result, not a new proof from the present run.

Original signal and exact question

Yoshikazu Giga and Xingfei Xiang, *Lorentz space estimates for vector fields with divergence and curl in Hardy spaces*, arXiv:1312.5823v1 [1], write on source-PDF page 1 that their work originates from Bourgain and Brezis’s Open Problem 1 and that they still do not know how to treat the case of a divergence-free field whose curl is merely in L^1 . Their own theorem instead assumes atomic-Hardy data on a bounded domain.

The cited problem is Bourgain–Brezis Open Problem 1: if $F \in L^1(\mathbb{T}^3; \mathbb{R}^3)$ is divergence-free and has mean zero, and Z is the mean-zero solution of

$$\operatorname{curl} Z = F, \quad \operatorname{div} Z = 0,$$

must Z belong to the critical Lorentz space $L^{3/2,1}(\mathbb{T}^3; \mathbb{R}^3)$? Equivalently, is there a uniform bound

$$\|Z\|_{L^{3/2,1}(\mathbb{T}^3)} \leq C\|F\|_{L^1(\mathbb{T}^3)}?$$

Decisive later answer

Jesse Goodman, Felipe Hernández, and Daniel Spector, *Potential Estimates and Hodge Systems with L^1 data on compact manifolds*, arXiv:2606.12891v1 [2], explicitly say in the abstract and Introduction that they answer Bourgain–Brezis Open Question 1.

Their Corollary 1.4 (source-PDF pages 4–5) proves exactly that, for every divergence-free mean-zero $F \in L^1(\mathbb{T}^3; \mathbb{R}^3)$, the canonical solution $Z = \operatorname{curl}(-\Delta_{\mathbb{T}^3})^{-1}F$ satisfies

$$\operatorname{curl} Z = F, \quad \operatorname{div} Z = 0, \quad \|Z\|_{L^{3/2,1}(\mathbb{T}^3; \mathbb{R}^3)} \leq C\|F\|_{L^1(\mathbb{T}^3; \mathbb{R}^3)}.$$

It also proves the particular consequence $Z(x)/|x - a| \in L^1(\mathbb{T}^3)$ for every $a \in \mathbb{T}^3$ asked in the original problem. Corollary 1.4 follows from their more general Theorem 1.3, an endpoint Lorentz estimate for Hodge systems on smooth compact Riemannian manifolds.

Identification, provenance, and scope

The match is exact and author-acknowledged: the later paper cites the original Bourgain–Brezis problem by number and page, states that it gives an affirmative answer, and its Corollary 1.4 has precisely the same hypotheses and conclusion. The run’s cheap indexes also already contained the [arXiv:2606.12891](#) extraction record, which led to this cross-check; a bounded search of the exact title, exact problem phrase, and the two arXiv identifiers confirmed the match.

The answer concerns the cited periodic problem on $\mathbb{T}^3$ (and the supporting paper substantially generalizes it to compact manifolds). It does not by itself assert every possible bounded-domain analogue with boundary conditions. However, [arXiv:1312.5823](#) does not formulate such an analogue as a separate numbered question: its open-problem signal expressly points to Bourgain–Brezis Open Problem 1. Thus that extracted target is fully answered. The packet stores the original paper as `source_paper.pdf` and the answer paper as `supporting_paper_2606.12891.pdf`.

References

- [1] Y. Giga and X. Xiang, *Lorentz space estimates for vector fields with divergence and curl in Hardy spaces*, Hokkaido University Preprint Series in Mathematics 1044 (2013), 1–11; [arXiv:1312.5823](#).
- [2] J. Goodman, F. Hernández, and D. Spector, *Potential Estimates and Hodge Systems with L^1 data on compact manifolds*, [arXiv:2606.12891v1](#) (2026), Theorem 1.3 and Corollary 1.4.